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Tier 1 Concept Grounding



Clinical text
“Heart attack”
“Necrosis of
the
myocardium
...”



Grounded biomedical concepts



Disease



Phenotype



Gene



Biomedical Knowledge Base



Expert-curated
definitions
UMLS, HPO, OMIM,
Gene Ontology, ...



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Tier 2 Ontology Relations

Inferred or curated relations

Phenotype → Disease



Nocturia

Gitelman
syndrome

Disease → Phenotype

Marfan
syndromeMitral
regurgitation

Scoliosis



Phenotype → Gene



Cyanosis



GABRG2

Disease → Gene

HYLS1
syndrome

HYLS1

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Tier 3 Compositional Retrieval

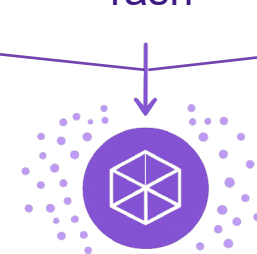
Input phenotype set



Fever

Skin
rashJoint
pain

...



Embedding
& evidence
aggregation



Top candidate diseases (ranked)

1	Lupus		0.87
2	Rheumatoid arthritis		0.72
3	Measles		0.61
4	Parvovirus B19 infection		0.43
5	...		...